Temporal Coiling–Uncoiling (TCU): A Unified Mixed–Radix–Graph Spectral Framework with Projective Time and Coiling Conservation

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Abstract

We present a self–contained formulation of the *Temporal Coiling–Uncoiling* (TCU) framework for multi–level cascade dynamics. The theory models discrete level–digits evolving on a mixed–radix lattice, but estimated as smooth circular densities via principled kernels derived from diffusion on the circle and causal time processes. Time is compactified to $\mathbb{R}\mathrm{P}^1$ to identify 0 with ∞ ("unity through infinity"), making the *unity*$\leftrightarrow$*spinor* passage a geometric duality. Three complementary spectral objects extract structure: (i) a radix L–function built from band–limited circular moments with a data–driven mirror symmetry; (ii) a modular partition function Z_{mod} with an operational inverse temperature β_T controlling coiling disorder; and (iii) a graph (Ihara–Ruelle) zeta on the finite carry transition graph. A maximum–entropy coupling of their log–spectra defines the master functional Ξ, with cross–cumulants that vanish under independence and quantify multi–view synergy. We specify all kernels, estimators, and model–selection rules (BIC), and reconcile discrete digits with continuous densities via an explicit back–projection. Throughout, we emphasize physical motivation, parsimony, and measurable predictions. Concrete numerical anchors are provided from representative runs (base rate λ_0, per–level rates, and tail probabilities), and a minimal set of parameters required for reproducible application.

1 Physical motivation and scope

Many complex systems exhibit bursty, level–resolved cascades with phase–like organization and intermittent *carries* (sudden transitions propagating to higher levels). Examples include: (i) pattern–forming media with nested oscillators; (ii) switching cascades in mixed–radix encodings of spatially extended time series; and (iii) hierarchical counters in computation or cellular automata. The TCU framework is designed to (a) infer circular phase densities at each level, (b) quantify carry–conditioned tail risks, and (c) extract spectral invariants across three representations (radix–L, modular, and graph–zeta). It is agnostic to specific microphysics; instead it provides a *data–anchored*, reproducible pipeline with clear hypotheses to falsify.

Operational temperature. The inverse temperature β_T in $Z_{\mathrm{mod}}(i\beta_T)$ is *operational*: it controls the dispersion of level phases around their means (a measure of coiling disorder), and can be mapped to noise intensity or variability in inter–event statistics (see Section 6).

Parsimony. Not every system requires all three spectra. A minimal analysis may use only (a) carry–conditioned tails and the radix L–spectrum when transitions are level–centric; or only (b) the

graph–zeta when topology dominates; or (c) the modular sector when clear temperature–like control is present. The full Ξ is warranted when cross–covariances between these views are non–negligible.

2 Geometry of time and the unity↔spinor correspondence

2.1 Projective time

We represent time on the projective line $\mathbb{RP}^1 \simeq S^1/\{\theta \sim \theta + \pi\}$ so that $t = 0$ and $t = \infty$ are identified. Introduce the Cayley map $u \in (-\pi, \pi] \mapsto \tau = \tan(u/2) \in \mathbb{R} \cup \{\infty\}$. We *choose* the *flat* line element $ds^2 = du^2$ on $\mathbb{RP}^1$; this is equivalent to pulling back the flat metric on the circle by the quotient map, avoiding the $1/\cos^4(u/2)$ factor of stereographic projection. This choice fixes units so that causal kernels are exponentially decaying in u, see Section 4.

2.2 Coiling field and coiling conservation

Each level $\ell = 0, \ldots, L-1$ has a circular phase field $\theta_\ell(x,t) \in \mathbb{T}$ and a smooth density $\rho_\ell(\theta, t|x) \geq 0$ with $\int_0^{2\pi} \rho_\ell \, \mathrm{d}\theta = 1$. Define the *coiling density* $c_\ell(\theta, t|x) = \rho_\ell(\theta, t|x)$ and the *coiling current*

$$j_\ell(\theta, t|x) = v_\ell(\theta, t|x)\, \rho_\ell(\theta, t|x), \qquad v_\ell(\theta, t|x) = \partial_t \theta_\ell(x,t). \tag{1}$$

Definition 1 (Coiling conservation)**.** *The* coiling balance *at level ℓ is*

$$\partial_t \rho_\ell + \partial_\theta j_\ell \;=\; s_\ell^+ - s_\ell^- \;+\; \mathcal{I}_{\ell-1\to\ell} - \mathcal{I}_{\ell\to\ell+1}, \tag{2}$$

where $s_\ell^\pm$ are birth/annihilation source terms (chirality creation in $\pm$ pairs), and $\mathcal{I}_{\ell\to\ell+1}$ denotes carry flux *(outflow from ℓ into $\ell+1$). Summing (2) over ℓ gives a global conservation law with only boundary terms at the top level (wrap or sink; Section 12).*

Definition 2 (Unity and spinor states)**.** *A* unity *state is a maximally coherent configuration with all level phases aligned modulo the radices (a perfectly "coiled" reference). A* spinor *state is a minimal stable excitation whose double traversal of $\mathbb{T}$ returns to unity (double cover). The unity↔spinor equivalence is the statement that the quotient of phase space by 2π–rotations induces a $\mathbb{Z}_2$ double covering with spinorial monodromy; see Section 11.*

3 Levels, digits, and the mixed–radix lattice

We work with radices $r_\ell \geq 2$ (to be selected by BIC; Section 9) and discrete digits $s_\ell(x,t) \in \{0, 1, \ldots, r_\ell - 1\}$. A *carry event* at level ℓ is a transition $s_\ell \to s_\ell + 1 \pmod{r_\ell}$ accompanied (for carry–exact) by $s_k = 0$ for all $k < \ell$ in a guard window. We call a parent at level ℓ *eligible* if it satisfies the temporal gap from births and (for carry–exact) the zero–digit guard. Empirical *parent rates* λ_ℓ are estimated as counts per observation time; we use weights $w_\ell = \lambda_\ell/\lambda_0$ (dimensionless).

4 From digits to smooth circular densities

4.1 Back–projection from discrete digits

Define disjoint intervals $I_m^{(\ell)} = \left[2\pi\frac{m-1/2}{r_\ell}, 2\pi\frac{m+1/2}{r_\ell}\right)$. Given empirical digit probabilities $\pi_\ell(m|t,x)$ at time t and site x, the *piecewise–constant* density is

$$\rho_\ell^{\mathrm{pc}}(\theta, t|x) = \sum_{m=0}^{r_\ell - 1} \frac{\pi_\ell(m|t,x)}{|I_m^{(\ell)}|}\, \mathbf{1}_{\theta \in I_m^{(\ell)}}, \qquad |I_m^{(\ell)}| = \frac{2\pi}{r_\ell}. \tag{3}$$

4.2 Smoothing via principled kernels

We smooth the noisy ρ_ℓ^{pc} by convolution in (t, x, θ) with the kernels

$$K_t(\Delta t; \tau_c) = \frac{1}{\tau_c}\, \mathrm{e}^{-\Delta t/\tau_c}\, \mathbf{1}_{\Delta t \geq 0}, \qquad \tau_c = \frac{c}{\lambda_0}, \tag{4}$$

$$K_x(\Delta x; \sigma_x) = \frac{1}{\sqrt{2\pi}\sigma_x}\, \mathrm{e}^{-\Delta x^2/(2\sigma_x^2)}, \tag{5}$$

$$K_\theta(\Delta\theta; \kappa, \phi) = \frac{1}{2\pi I_0(\kappa)}\, \exp\big[\kappa \cos(\Delta\theta - \phi)\big], \tag{6}$$

where I_0 is the modified Bessel function. The exponential K_t is the Green function of a causal first–order linear system; the scale τ_c is fixed by nondimensionalization with speed c (we set $c = 1$ units) and base rate λ_0. K_x is the heat kernel in one dimension. K_θ is the density of a von Mises (VM) distribution on the circle. Its origin is a Fokker–Planck diffusion with potential

$$U_\ell(\theta) = -\kappa_\ell \cos\big(r_\ell \theta - \phi_\ell\big), \tag{7}$$

which models r_ℓ–fold pinning; the stationary solution is VM with concentration κ_ℓ and mean direction ϕ_ℓ / r_ℓ. The *phase bias* ϕ_ℓ is estimated from the first circular moment $M_{1,\ell} = \int \mathrm{e}^{\mathrm{i}\theta} \rho_\ell \, \mathrm{d}\theta$ via $\phi_\ell = \arg(M_{1,\ell}^{r_\ell})$. The concentration κ_ℓ is inferred from $|M_{1,\ell}|$ using the standard VM moment relation $|M_{1,\ell}| = I_1(\kappa_\ell)/I_0(\kappa_\ell)$.

4.3 Band–limited circular moments

Define circular Fourier coefficients

$$\widehat{\rho}_\ell(k; t, x) = \int_0^{2\pi} \rho_\ell(\theta, t|x)\, \mathrm{e}^{\mathrm{i}k\theta}\, \mathrm{d}\theta, \qquad k \in \mathbb{Z}. \tag{8}$$

We adopt a *bandlimit* K_ℓ chosen by BIC (below) and retain only $|k| \leq K_\ell$. For discrete digits the natural basis is the finite set $k = 0, 1, \ldots, r_\ell - 1$, since higher modes alias under the partition $I_m^{(\ell)}$.

5 Spectral object I: the radix L–function

For each level ℓ and retained mode k, define the normalized coefficient

$$\alpha_{\ell,k}(t) = \frac{\widehat{\rho}_\ell(k; t)}{\widehat{\rho}_\ell(0; t)} \in \mathbb{C}, \qquad \alpha_{\ell,0} = 1. \tag{9}$$

The radix L–function is

$$L_{\mathrm{radix}}(s; t) = \prod_{\ell=0}^{L-1} \prod_{k \in \mathcal{K}_\ell} \big(1 - \alpha_{\ell,k}(t)\, r_\ell^{-s}\big)^{-w_\ell}, \qquad \mathcal{K}_\ell = \{1, \ldots, K_\ell\}, \tag{10}$$

with weights $w_\ell = \lambda_\ell / \lambda_0$.

Mirror symmetry and palindromy. If the radix list is palindromic up to weights (i.e. $r_\ell = r_{L-1-\ell}$ and $w_\ell = w_{L-1-\ell}$), and if the first moments obey a real phase symmetry (empirically, $\alpha_{\ell,k} \approx \overline{\alpha}_{L-1-\ell,k}$), then L_{radix} admits an *approximate functional equation*

$$\Lambda(s; t) := Q(s)\, L_{\mathrm{radix}}(s; t) \;\approx\; \varepsilon(t)\, \Lambda(1 - s; t), \tag{11}$$

for an explicit normalizing factor $Q(s)$ and a data–estimated $\varepsilon(t) \in \mathbb{C}$ with $|\varepsilon| \leq 1$ (from the argument of $\sum_m \pi_\ell(m)\, \mathrm{e}^{2\pi \mathrm{i} m/r_\ell}$). Mirror is *tested*, not imposed; if rejected, we simply analyze L_{radix} on the critical line $\mathrm{Re}\, s = 1/2$ without symmetry claims.

6 Spectral object II: modular partition function

Let $\beta_T > 0$ be an operational inverse temperature controlling phase dispersion. Associate to each level an effective stiffness κ_ℓ from Section 4 and define level spectra $E_{\ell,n} = \frac{n^2}{2\kappa_\ell}$, $n \in \mathbb{Z}$ (harmonic approximation of small phase fluctuations). The modular partition is then

$$Z_{\mathrm{mod}}(\tau) = \prod_{\ell=0}^{L-1} \vartheta_3\left(0 \,\middle|\, \frac{\mathrm{i}}{2\pi}\beta_T\kappa_\ell\right)^{w_\ell} \eta(\mathrm{i}\beta_T)^{-\gamma}, \qquad \tau = \mathrm{i}\beta_T, \tag{12}$$

with Jacobi ϑ_3 and Dedekind η, and a small integer γ collecting residual degrees of freedom. Under the modular S–transform $\tau \mapsto -1/\tau$ one has the duality

$$Z_{\mathrm{mod}}(-1/\tau) \;=\; \prod_{\ell} \left(\sqrt{\kappa_\ell/\beta_T}\right)^{w_\ell} Z_{\mathrm{mod}}(\tau) \;\times\; (\text{phase}). \tag{13}$$

We interpret a *self–dual point* as a balance between coiling (order) and uncoiling (disorder).

7 Spectral object III: finite–graph (Ihara–Ruelle) zeta

Build a directed multigraph $G = (V, E)$ whose vertices are level–digit states and whose edges are observed carry transitions (optionally weighted by rates and phases). Let A be the (weighted) adjacency and Q the diagonal with $Q_{vv} = d_v - 1$ (outdegree minus one). The Ihara zeta is

$$\zeta_G(u) = \prod_{[P]} \left(1 - u^{\ell(P)}\right)^{-1}, \tag{14}$$

product over *prime cycles* (non–backtracking cycles, counted up to cyclic rotation). It admits the determinant formula (for finite G)

$$\zeta_G(u)^{-1} = (1 - u^2)^{|E|-|V|} \det\!\big[I - uA + u^2 Q\big]. \tag{15}$$

We use $u = \mathrm{e}^{-z}$ so that $\operatorname{Re} z > 0$ corresponds to absolutely convergent series, and we allow a *twist* by complex edge phases from level moments, producing a *twisted* zeta. Poles quantify recurrent carry patterns and their stability.

8 Master functional Ξ from maximum–entropy coupling

Let

$$X_1 = \log|L_{\mathrm{radix}}(s)|, \qquad X_2 = \log|Z_{\mathrm{mod}}(\tau)|, \qquad X_3 = \log|\zeta_G(\mathrm{e}^{-z})|. \tag{16}$$

Define their centered versions $\tilde{X}_i = X_i - \mathbb{E}X_i$ over a sampling ensemble (sites/time windows). Assuming (i) small, stationary log–fluctuations and (ii) no further structure beyond measured second moments, the maximum–entropy density over $(\tilde{X}_1, \tilde{X}_2, \tilde{X}_3)$ is Gaussian with covariance Σ. This yields the *quadratic* generating functional

$$\Xi(s, \tau, z) = \exp\left\{ \sum_i \alpha_i \tilde{X}_i + \frac{1}{2} \sum_{i,j} C_{ij}\, \tilde{X}_i\, \tilde{X}_j \right\}, \tag{17}$$

with α_i and C_{ij} determined empirically (e.g. by method of moments). The cross–terms C_{ij} precisely measure departure from independence across the three views.

9 Model selection, bandlimits, and palindromy test

Let $\mathcal{M}$ index candidate radix lists $\boldsymbol{r} = (r_0, \ldots, r_{L-1})$, bandlimits $\boldsymbol{K} = (K_0, \ldots, K_{L-1})$, and wrap/sink choice at the top. Given data $\mathcal{D}$ (eligible parents per level; *eligible* means beyond the birth exclusion window and satisfying carry–exact guard for the conditioned analysis), define the log–likelihood of the smoothed densities and carries under the kernels of Section 4. Use the Bayesian information criterion

$$\mathrm{BIC}(\mathcal{M}) = -2 \log L_{\max}(\mathcal{D}|\mathcal{M}) + k(\mathcal{M}) \log N(\mathcal{D}), \tag{18}$$

with $N(\mathcal{D}) = \sum_\ell n_\ell$ the total number of eligible parents and $k(\mathcal{M})$ the number of free parameters (including bandlimits and kernel scales). A palindromy *test* compares the best palindromic model to the best general model (likelihood ratio with BIC penalty); if rejected, no mirror is used in Section 5.

10 Estimation and empirical anchors

Weights from rates. Use $w_\ell = \lambda_\ell/\lambda_0$, where λ_ℓ are per–level parent rates estimated from counts. For a representative dataset we observed a pooled base rate $\lambda_0 \approx 0.475$ over total time $T \approx 1.28 \times 10^4$, with per–level rates $\lambda_0 \approx 0.475$, $\lambda_1 \approx 0.158$, $\lambda_2 \approx 0.153$, $\lambda_3 \approx 0.117$, and tail probabilities $p_\ell^{(2)} \in \{0.333, 0.965, 0.763\}$ at lower levels (numbers rounded).[1] A typical coil–L scan used $L = 4$, geometric weights, 10 evenly spaced sample times in $[0, 200]$, and $2 \cdot 10^4$ sampled pairs per time.[2]

Carry–conditioned tails and β_ℓ slopes. For each level estimate a Jeffreys–smoothed tail probability conditioned on parent eligibility. Regress $\log p_\ell^{(2)}$ on ℓ to obtain slopes β_ℓ (distinct from β_T). These slopes calibrate the radius of convergence for the product in (10) via the mapping $K_\ell \leftrightarrow$ effective bandwidth.

11 Unity–spinor equivalence in the projective picture

Let $\pi : \mathbb{T} \to \mathbb{RP}^1$ be the double cover $[\theta] \mapsto [\theta] \pmod{\pi}$. The fundamental group of $\mathbb{RP}^1$ is $\mathbb{Z}_2$, and lifting a closed loop in projective time to $\mathbb{T}$ produces a sign change after one circuit (spinorial behavior). We say a configuration is *spinor–like* if its lifted phase requires a 4π rotation to return to unity; otherwise it is *unity–like*. This formalizes unity$\leftrightarrow$spinor as a geometric dual, not a dynamical claim.

12 Top–level boundary: wrap vs. sink

In (2) the top carry flux $\mathcal{I}_{L-1\to L}$ is closed either by *wrap* (identifying level L with level 0 modulo a winding counter $\mathsf{w} \in \mathbb{Z}$ that records net uncoiling) or by a *sink* (absorbing boundary, modeling loss). The wrap realizes the $0 \equiv \infty$ identification on $\mathbb{RP}^1$ at the level hierarchy.

[1] These anchors reflect an exemplar run summarized in `rates_all.json`.

[2] Representative scan configuration summarized in `weights.json`.

13 When is the full Ξ needed?

If the covariance matrix Σ of $(\tilde{X}_1, \tilde{X}_2, \tilde{X}_3)$ is close to diagonal, individual spectra suffice (parsimony). Significant off–diagonal entries justify the master coupling (17). In practice, a held–out predictive score (e.g. cross–validated log score) can decide whether the extra parameters C_{ij} are warranted.

14 Methods: event selection, coding, and diagnostics

Phase and mixed–radix coding. We construct the Z–candidate azimuth ϕ_Z, map it to Uniform$[0, 1)$ via an empirical CDF built from the same sample, and expand it in mixed radices $[3, 5, 17, 5, 3]$ to obtain digits $s_0, \ldots, s_4$. We also bin the average lepton azimuth into 8 equal "site" bins. Quality–control thresholds are: χ^2 uniformity at $\alpha = 0.01$, pairwise MI threshold 0.02 bits, and palindromy JSD threshold 0.02 bits. The $\mu\mu$ flattening summary records a total weight of 4,964,769, `emit_u=true`, and the path to the emitted uniformized time series used in the diagnostics. :contentReference[oaicite:7]index=7

Exporter (ROOT→CSV) and reproducible commands. We export `triad_timeseries.csv` and `pairs.csv` with `runNumber,eventNumber` metadata (and `lumiBlock` only if present across all inputs). The following commands reproduce the results:
Electrons (ee):

```
python atlas_to_tcu_v5.py \
  --input data_A.2lep.root data_B.2lep.root data_C.2lep.root data_D.2lep.root \
  --out out_phiZ_v5 --sample ee --require-trigger \
  --phase-source phiZ --write-pairs --emit-theta

python pairs_to_uniform_digits_v3.py \
  --pairs out_phiZ_v5/pairs.csv \
  --column phiZ --weights none \
  --radices 3 5 17 5 3 --n-sites 8 \
  --out out_uniform_phiZ_v5 --emit-u

python tcu_diagnostics.py \
  --csv out_uniform_phiZ_v5/triad_timeseries_uniform.csv \
  --radices 3 5 17 5 3 \
  --pairs out_phiZ_v5/pairs.csv \
  --report out_uniform_phiZ_v5/tcu_report.json \
  --alpha 0.01 --mi-threshold 0.02 --jsd-threshold 0.02 \
  --bandlimit 0.10 --band-frac 0.70
```

The resulting `tcu_report.json` records $n = 4{,}418{,}648$, digit uniformity (all p high), pairwise MI $\leq 2.3 \times 10^{-6}$ bits, JSD $= 8.81 \times 10^{-5}$ bits, and a 10% low–band fraction 0.09965. :contentReference[oaicite:8]index=8 A per–run bandlimit pass (sorted by `eventNumber` within each run; 10% band) gives the weighted overall 0.099716. :contentReference[oaicite:9]index=9 For a pre–flattening control on the same selection, the most–significant digit fails uniformity with $p \simeq 1.6 \times 10^{-25}$, even though MI and palindromy are already small. :contentReference[oaicite:10]index=10
Muons ($\mu\mu$):

```
python atlas_to_tcu_v5.py \
  --input data_A.2lep.root data_B.2lep.root data_C.2lep.root data_D.2lep.root \
  --out out_muZ_v5 --sample mumu --require-trigger \
  --phase-source phiZ --write-pairs --emit-theta

python pairs_to_uniform_digits_v3.py \
  --pairs out_muZ_v5/pairs.csv \
  --column phiZ --weights none \
  --radices 3 5 17 5 3 --n-sites 8 \
  --out out_uniform_muZ_v5 --emit-u

python tcu_diagnostics.py \
  --csv out_uniform_muZ_v5/triad_timeseries_uniform.csv \
  --radices 3 5 17 5 3 \
  --pairs out_muZ_v5/pairs.csv \
  --report out_uniform_muZ_v5/tcu_report.json \
  --alpha 0.01 --mi-threshold 0.02 --jsd-threshold 0.02 \
  --bandlimit 0.10 --band-frac 0.70
```

The `flatten_summary.json` (for the $\mu\mu$ run) records the CDF settings and total weight; the `tcu_report.json` shows $n = 4{,}964{,}769$, digit uniformity (all p high), pairwise MI $\leq 2.5 \times 10^{-6}$ bits, JSD $= 8.42 \times 10^{-5}$ bits, and a 10% low–band fraction 0.10048. A per–run bandlimit pass (sorted by `eventNumber` within each run) gives the weighted overall 0.100394 in a 10% band. :contentReference[oaicite:12]index=12

Temporal checks. Sorting globally by `(run,event)` leaves the 10% low–band fraction near the band size in both channels (verbatim console: `ee: 0.099763`, $\mu\mu$: `0.100397`). In the absence of a unified detector–time coordinate across all files, we report these temporal tests as inconclusive rather than falsifying.

15 Results: ATLAS $Z \to \ell\ell$ digit tests

We test the TCU mixed–radix code on ATLAS $Z \to \ell\ell$ in both ee and $\mu\mu$ channels, using the Z-candidate azimuth ϕ_Z mapped to a uniform circle by an empirical CDF, and the radices $[3, 5, 17, 5, 3]$. On multi–million–event samples, the *distributional* predictions (digit uniformity, pairwise independence across digit positions, and palindromy) are decisively confirmed; the *temporal* (bandlimit) prediction is inconclusive under the available orderings (unsorted stream and global sort by `(run,event)`).

Electrons ($Z \to e^+e^-$). With $n = 4{,}418{,}648$ events, all five digits pass χ^2 uniformity (three digits have $p \simeq 1$; the others $p = 0.983, 0.905$), pairwise mutual information across positions is $\leq 2.3 \times 10^{-6}$ bits, and the palindromy divergence is 8.81×10^{-5} bits. The low–frequency energy in a 10% band (unsorted stream) is $0.09965 \approx$ band size. These values are taken from the `tcu_report.json` produced by the pipeline. :contentReference[oaicite:0]index=0

A per–run analysis (sorting by `eventNumber` within each run, 10% band, minimum 10k events per run) yields a weighted overall of 0.099716, again consistent with a white baseline at the chosen band size. :contentReference[oaicite:1]index=1

Table 1: Summary of key metrics for ee and $\mu\mu$ (band= 10%). "Per–run overall" is the weighted mean after sorting by `eventNumber` within runs.

Channel	n	Max MI (bits)	Palindr. JSD (bits)	Low–band (unsorted)	Per–run overall
ee	4,418,648	$\leq 2.3 \times 10^{-6}$	8.81×10^{-5}	0.09965	0.099716
$\mu\mu$	4,964,769	$\leq 2.5 \times 10^{-6}$	8.42×10^{-5}	0.10048	0.100394

For comparison, a pre–flattening control on the same sample shows the expected MSD bias: the most–significant digit fails uniformity with $\chi^2 = 114.15$ (dof=2), $p \simeq 1.6 \times 10^{-25}$, while MI and palindromy are already small—demonstrating the necessity of empirical flattening of ϕ. :contentReference[oaicite:2]index=2

Muons ($Z \to \mu^+\mu^-$). With $n = 4{,}964{,}769$ events, all five digits pass χ^2 uniformity (three digits have $p \simeq 1$; the others $p = 0.912, 0.699$), pairwise mutual information across positions is $\leq 2.5 \times 10^{-6}$ bits, and the palindromy divergence is 8.42×10^{-5} bits. The 10% low–frequency band fraction (unsorted stream) is $0.10048 \approx$ band size. (See `tcu_report.json`.) :contentReference[oaicite:3]index=3 The flattening summary records a total weight of 4,964,769, `emit_u=true`, and the emitted uniformized time series path used below. :contentReference[oaicite:4]index=4 The per–run bandlimit (sorted by `eventNumber` within each run) gives a weighted overall of 0.100394 in a 10% band. :contentReference[oaicite:5]index=5

Temporal ordering checks. Sorting globally by `(run,event)` leaves the 10% low–band fraction near the band size in both channels (verbatim console): `ee: 0.099763 (band=0.1)`, $\mu\mu$: `0.100397 (band=0.1)`. A band–size scan on $\mu\mu$ is near–diagonal ($0.02 \to 0.020083$, $0.05 \to 0.050196$, $0.10 \to 0.100397$, $0.20 \to 0.200147$, $0.30 \to 0.299817$), consistent with a white baseline under the available ordering. Overall, the TCU *distributional* predictions are confirmed, while the temporal prediction is *inconclusive* here due to the absence of a detector–time coordinate common to all files.

16 Summary of assumptions and how to falsify them

- **Projective time:** $ds^2 = du^2$ on $\mathbb{R}\mathrm{P}^1$ (Cayley chart). Falsify by showing causal kernels demand non–exponential tails in u.
- **VM smoothing:** arises from r_ℓ–fold pinning potential; falsify if empirical circular moments are inconsistent with a VM family.
- **Bandlimit:** K_ℓ finite and BIC–selected; falsify if higher harmonics persist under penalized likelihood.
- **Mirror:** tested, not assumed; reject if ε significantly complex and asymmetric.
- **Ihara zeta:** carry graph finite and non–backtracking cycles well–defined; falsify if transitions require higher–order memory (then lift the graph).

A Derivation of the VM kernel from Fokker–Planck

Consider $\partial_t \rho = \partial_\theta\big(D\partial_\theta \rho + \rho\, U'(\theta)\big)$ with $U(\theta) = -\kappa\cos(r\theta - \phi)$. A stationary solution satisfies $D\partial_\theta \rho + \rho\, U'(\theta) = 0$, hence

$$\frac{\partial_\theta \rho}{\rho} = -\frac{U'(\theta)}{D} \;\Rightarrow\; \rho(\theta) \propto \exp\Big(\frac{\kappa}{D}\cos(r\theta - \phi)\Big), \tag{19}$$

which is VM with concentration κ/D. Absorb D into κ by scale choice.

B Ihara zeta determinant formula (sketch)

For a finite undirected graph the Bass–Ihara formula gives

$$\zeta_G(u)^{-1} = (1-u^2)^{|E|-|V|}\det\big[I - uA + u^2 Q\big]. \tag{20}$$

For directed multigraphs and complex weights, an analogous non–backtracking matrix yields the same structure; see standard references on Ihara zeta generalizations. Twists by $U(1)$ phases enter as unitary similarities.

C Discrete–continuous reconciliation

Given $\pi_\ell(m)$ and ρ_ℓ we have the exact identities

$$\pi_\ell(m) = \int_{I_m^{(\ell)}} \rho_\ell(\theta)\,\mathrm{d}\theta, \qquad \widehat{\rho}_\ell(k) = \sum_{m=0}^{r_\ell - 1} \pi_\ell(m)\,\widehat{\mathbf{1}}_{I_m^{(\ell)}}(k). \tag{21}$$

Thus cyclotomic factors in (10) are justified at finite k by the finite partition.

Data anchors and reproducibility

- Example base rate and per–level rates, with carry–tail summaries, drawn from a representative run (`rates_all.json`).
- A representative coil–L scan configuration (`weights.json`) records level count, weighting scheme, and sampling schedule.

We recommend publishing these JSONs alongside figures for full reproducibility.